ExamDesk

Online Examination System

A cloud-hosted platform for conducting exams and delivering results

Project Presentation

Introduction

- ExamDesk is a web-based system for taking exams online.
- Students can register, log in, view available exams and answer questions.
- The backend checks the submitted answers and saves the result in a database.
- Frontend and backend run on separate AWS EC2 instances, with Amazon RDS for data storage.

Goal: a clear, centralized and easily accessible online exam process.

Project Objectives

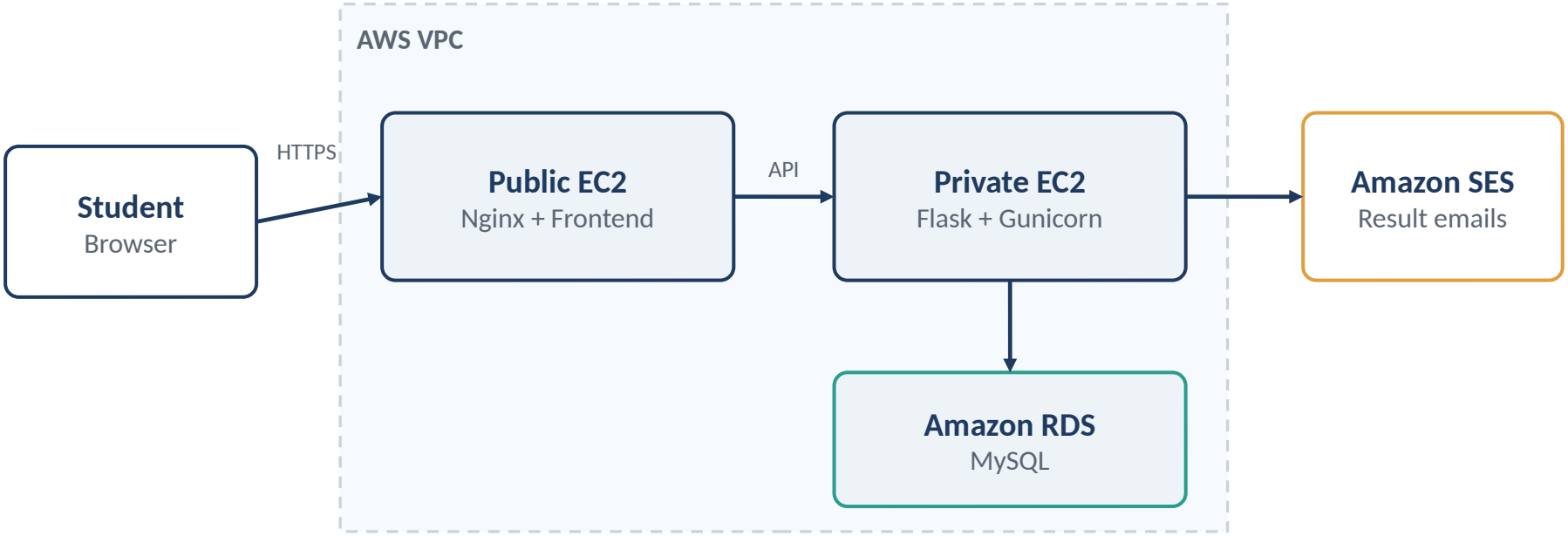
1. **Online access** — A web interface for students to access exams.
2. **Secure login** — Authenticate users before they use protected features.
3. **Reliable submission** — Receive answers and process submitted exam attempts.
4. **Result storage** — Store attempts and results in MySQL on Amazon RDS.
5. **Email notification** — Send result emails using Amazon SES.

Proposed System

Student Portal	Application Backend	Data & Cloud Services
• Register and log in • Browse available exams • Answer questions and submit • View the result	• Validate requests and login • Load exam questions • Calculate scores • Trigger result email	• RDS (MySQL) stores data • EC2 hosts the application • VPC and IAM for security • SES sends emails

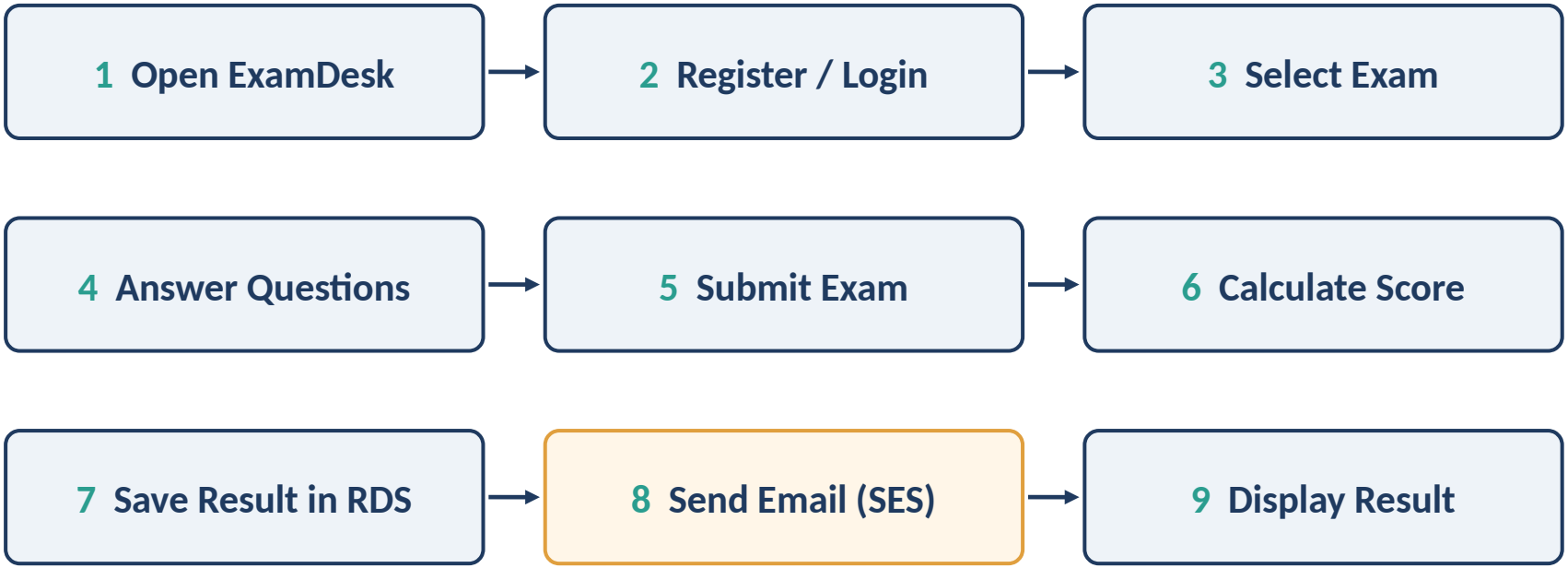
Flow: Student interface → Application logic → Database and cloud services

System Architecture



The frontend and backend run on separate EC2 instances. Only the private backend connects to the database.

Student Exam Workflow



The score is calculated and saved first; the email is attempted afterwards.

Note: SES is in sandbox mode, so recipient emails must be verified.

AWS Services Used

Service	Purpose in ExamDesk
Amazon EC2	Hosts the frontend and Flask backend on separate instances.
Amazon RDS	Managed MySQL database for users, exams, attempts and results.
Amazon VPC	Network isolation for public and private resources.
AWS IAM	Controls service permissions through roles and policies.
Amazon SES	Sends exam result emails.
Amazon S3	Object storage used in the project environment.

Database and Security

Database

- Amazon RDS with MySQL
- Stores user and exam data
- Stores submitted attempts and results
- Backend connects to it for all data operations

Security

- Login required for protected routes
- Private backend is not publicly accessible
- VPC and security groups restrict network access
- IAM role grants AWS service permissions

Good practice: keep credentials in protected configuration, limit inbound rules and keep tested backups.

Conclusion

ExamDesk brings the examination process online. It combines a web frontend, Flask backend, MySQL storage and AWS infrastructure to support exam access, submission, result processing and email notification.

Future Enhancements

- Admin dashboard improvements
- More detailed reports
- Stronger automated testing
- Production email access